



PROFESSIONAL SUMMARY

Computer Science undergraduate at BITS Pilani with experience building AI-driven workflow systems and API-based backend services in Python. Developed LangGraph-based RAG pipelines with structured output validation and evaluation workflows. Working knowledge of SQL and distributed system fundamentals, with interest in designing reliable, scalable SaaS applications.

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani

2023 – 2027

*Bachelor of Engineering in Computer Science (Honors)**Pilani, India*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems, Object-Oriented Programming, Machine Learning, Computer Architecture, Design and Analysis of Algorithms

PROJECTS & RESEARCH EXPERIENCE

Finite Element–Based Neural Twin for Structural Dynamics and SHM | Python, MATLAB, Finite Element Analysis, Signal Processing

2025

- * Architected a physics-informed digital twin fusing FE simulations with deep neural networks over 10,000+ structural samples.
- * Engineered virtual sensing at uninstrumented locations via sparse boundary-constrained local–global–local reconstruction.
- * Automated model calibration using Bayesian surrogate optimization, cutting expensive FE evaluations significantly.
- * Achieved AUC up to 0.99 on benchmark damage detection; published in The Journal of Supercomputing (Springer, 2026).

Federated Learning with Flower and ALBERT for Distributed NLP | PyTorch, Flower (FLWR), ALBERT, Distributed Systems

2025

- * Deployed a privacy-preserving federated NLP system training ALBERT across 10+ distributed clients without data sharing.
- * Engineered communication-efficient aggregation logic, analyzing bandwidth overhead and convergence across client topologies.
- * Stress-tested the system under non-IID data distributions, diagnosing heterogeneity drift and applying corrective strategies.
- * Benchmarked scalability and training stability across client configurations, producing actionable optimization recommendations.

RLBenchKit - Reinforcement Learning Benchmarking Framework | Python, PyTorch, Gymnasium, PPO, DQN, Continuous Control

2025

- * Built a plug-and-play RL benchmarking suite enabling systematic algorithm comparison across diverse Gymnasium environments.
- * Designed automated evaluation pipelines tracking convergence curves, reward variance, and training stability metrics.
- * Uncovered cross-algorithm performance patterns through rigorous empirical analysis and interactive visualization dashboards.
- * Modeled experiment orchestration after production job schedulers, enabling scalable and reproducible research workflows.

Multilingual Child-Specific Content Safety LLM | NLP, FastAPI, Transformers, Multi-label Classification, Safety AI, Ollama

2026

- * Built a production-grade multilingual content moderation system detecting child-unsafe text across 10+ harm categories.
- * Designed an end-to-end multi-label NLP pipeline handling class imbalance, tokenization, and cross-lingual feature extraction.
- * Deployed low-latency FastAPI inference endpoints enabling real-time content moderation for downstream application integration.
- * Drove iterative model improvement through precision, recall, F1 analysis and hypothesis-driven error diagnosis.

PUBLICATIONS & CERTIFICATIONS

- **Research Publication – The Journal of Supercomputing, Springer Nature (2026):** Co-author, "Integrated Finite Element-Based Digital Twin Model and Scalable Computational Feature Transfer Framework for Structural Dynamics and Damage Detection," vol. 82, article 440. DOI: [10.1007/s11227-026-08578-3](https://doi.org/10.1007/s11227-026-08578-3)
- **AI Internship - Summer of AI (Swecha Tech)** - Worked on backend-oriented AI/ML applications. Built data processing pipelines, collaborated in cross-functional teams, and contributed to reliable, production-focused software development using Python and open-source tools.
- **TATA Cybersecurity Analyst Micro-Internship** - IAM security monitoring, and policy-based access.
- **Machine Learning Certification – Python & scikit-learn:** Model training and evaluation using Python and scikit-learn.

TECHNICAL SKILLS

Programming: Python, Java, C, SQL

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, System Design, REST APIs, Debugging, Testing

Backend & Systems: REST APIs, Client–Server Architecture, API Integration, Distributed Systems, Multithreading, System Design, Debugging, Testing

AI Workflow & LLM Systems: LangGraph, Retrieval-Augmented Generation (RAG), Structured Output Validation, Prompt Evaluation, Routing Logic, Retry & Fallback Handling

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, NumPy, Pandas, SciPy, FLWR, OpenCV, Librosa

Cloud & DevOps: Docker, Git, GitHub, Jupyter, Google Colab, MATLAB

LEADERSHIP & ACTIVITIES

- **Joint Secretary, Department of Firewallz:** Coordinated events and cross-team logistics; managed stakeholder communication.
- **Team Lead, SOAI Hackathon 2025:** Led a multidisciplinary team to deliver an AI solution within a 36-hour hackathon sprint.
- **Project Collaboration:** Collaborated with technical and non-technical stakeholders across multiple projects, contributing hands-on while supporting coordination and timely delivery.